



Introduction to Linux OS

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Creating a Collection of Files

If you want to bundle a group of files and/or directories together in an archive (not compressing), you can use the **tar** command which is one of the oldest and common methods for creating and working with compressed backup archives.

- You may want to create a copy or backup of a group of files.
- You may have several files you want to transfer at once or as a set.

```
$ tar [-] c|x|t f <tarfile> <pattern>
```

Note that you have the choice to precede the tar arguments with a hyphen (-) or omit it. For example, `$ tar cf file.tar` is the same as `$ tar -cf file.tar`.

tar option	Description
c	Create a tar archive.
f <file>	The tar archive file to perform operations against.
v	Causes tar to be verbose.
t	Display the table of contents (list).
x	Extract files from the archive.

Examples
<pre>\$ cd /home/MG \$ mkdir data \$ touch data/file.txt data/report.docx main.py \$ tar -cf archived.tar main.py data/ \$ tar -tf archived.tar \$ file archived.tar</pre>
<pre>\$ mkdir /tmp/extract \$ tar -xf archived.tar -C /tmp/extract \$ ls /tmp/extract/data/</pre>
<pre>\$ touch secret secret.bak \$ tar -cvf misc.tar sec* data \$ ls -lh</pre>



Compressing Files To Save Space

Compressing files in Linux is a common way to save space on disk. Linux provides several tools and utilities to compress files and directories effectively.

Command	Description
\$ gzip <file>	Compress file. The resulting compressed file is named file.gz.
\$ gunzip <file>	Uncompress files.
\$ gzcat or zcat	Concatenates and prints compressed files.

Hint: You can use the command **du** to display how much space is used by the file.

```
$ du -h archived.tar
$ gzip archived.tar
$ ls -lh
$ du -h archived.tar.gz
$ gunzip archived.tar.gz
$ ls -lh
```

Linux provides several other tools and utilities to compress files and directories effectively.

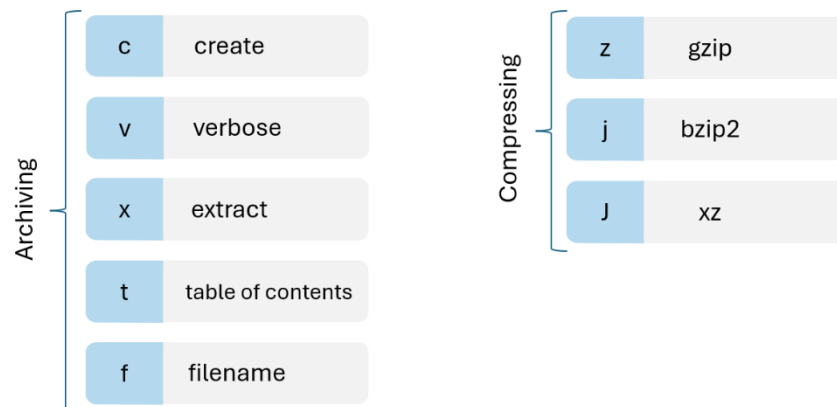
Compression	Decompression
\$ bzip2 <file>	\$ bunzip2 <file>
\$ zip <file>	\$ unzip <file>
\$ xz <file>	\$ xz <file>



Compressing Archives using tar

In modern versions of the **tar** command **gzip** , **bzip2**, and **xz** compressions are built-in.

Type	Compression Percentage	Extension	Compression Command
gzip	75%	.tar.gz	[root@centos9 ~]# tar zcfv file.tar.gz /mnt
bzip2	80%	.tar.bz2	[root@centos9 ~]# tar jcfv file.tar.bz2 /mnt
xz	85%	.tar.xz	[root@centos9 ~]# tar Jcfv file.tar.xz /mnt

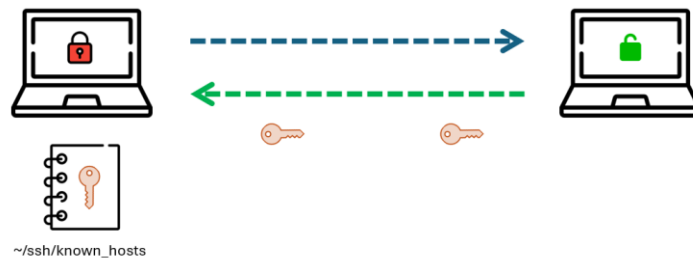


```
$ tar -zcvf compressed_gzip.tar.gz data
$ tar -jcvf compressed_bzip2.tar.bz2 data
$ tar -Jcvf compressed_xz.tar.xz data
```



Connecting remotely to other machines using SSH in Linux?

It's a protocol used to securely run a shell on a remote system. If you have a user account on a remote Linux System providing SSH services, you can remotely log into and run commands on that system.



Command	Usage
<code>sudo systemctl start sshd</code>	Start the SSH service on the system using systemctl.
<code>sudo systemctl stop sshd</code>	Stop the SSH service on a system using systemctl.
<code>sudo systemctl status sshd</code>	Display the status of the SSH service using the service command.
<code>whoami</code>	Display the username of the current user.
<code>ip a</code>	Display all IP addresses and network interfaces.
<code>ssh username@ip_address</code>	Connect to a remote machine using SSH with a specified username and IP address.
<code>exit</code>	Exit the current shell or terminates the SSH session.



Copying Files between Hosts

To copy files between hosts in a Linux environment, you can use several tools, with **scp**, **rsync**, and **sftp** being the most common.

scp	Usage	Strength
	Copy files securely over SSH	Simple, secure, works with individual files
	\$ scp file user@host:/path/to/destination	
rsync	Usage	Strength
	Copy files securely with efficient synchronization	Fast, supports incremental copying
	\$ rsync -avz file user@host:/path/to/destination	
sftp	Usage	Strength
	Interactive file transfer over SSH	Allows navigation and file management
	\$ sftp user@host	

Note: If you are running Windows, you can use the [PuTTY Secure Copy Client](#) (pscp.exe) and the [PuTTY Secure File Transfer](#) client (psftp.exe) programs.

Using scp (Secure Copy)

scp is a command-line tool for securely copying files between local and remote hosts over SSH.

```
$ scp [options] source_file username@destination_host:destination_path
```

▪ Copying a file from local to remote:

```
$ scp /path/to/local/file username@remote_host_ip:/path/to/remote/directory
```

```
$ scp /C/Users/MG/Downloads/file.txt MG@172.26.107.24:/home/MG
```

▪ Copying a file from remote to local:

```
$ scp username@remote_host:/path/to/remote/file /path/to/local/directory
```

```
$ scp MG@172.26.107.24:/home/MG/data.csv /C/Users/MG/Downloads
```



- **Copying a directory recursively (-r flag):**

```
$ scp -r /path/to/local/directory username@remote_host:/path/to/remote/directory  
$ scp -r username@remote_host:/path/to/remote/directory /path/to/local/directory
```

\$ scp -r /C/Users/MG/Downloads/Test MG@172.26.107.24:/home/MG
--

\$ scp -r MG@172.26.107.24:/home/MG/Pictures /C/Users/MG/Downloads
--

- **Using a different SSH port (default port is 22):**

```
$ scp -P port /path/to/local/file username@remote_host:/path/to/remote/directory
```

\$ scp -P 22 MG@172.26.107.24:/home/MG/data.csv /C/Users/MG/Downloads

Other Similar Commands:

- [rsync](#)
- [sftp](#)